

STUTTGART SCHNARREN, Germany

WMO: 107390

Lat: 48.833N

Lon: 9.200E

Elev: 315

StdP: 97.60

Time Zone: 1.00 (EUC)

Period: 84-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.5	-9.0	-14.7	1.1	-10.0	-12.2	1.4	-8.3	11.6	5.7	9.7	5.0	2.9	110	0.509

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.3	29.6	19.6	27.8	18.6	26.1	17.9	20.6	27.3	19.7	25.9	18.8	24.4	3.0	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.6	23.2	17.4	12.9	22.4	16.6	12.3	21.6	60.4	27.5	57.5	25.8	54.5	24.5	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.1	7.7	6.6	-13.1	33.1	5.4	1.4	-16.9	34.1	-20.1	35.0	-23.1	35.7	-27.0	36.8		
			-13.5	22.0	5.2	0.6	-17.2	22.4	-20.3	22.8	-23.3	23.1	-27.1	23.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	1.0	1.7	6.1	8.9	14.2	16.7	19.2	19.4	15.2	10.8	5.0	2.8
	DBStd	7.62	5.22	5.39	4.21	3.57	3.75	3.80	3.02	3.50	3.31	3.71	3.98	4.15
	HDD10.0	1131	279	233	132	62	10	1	0	0	1	35	153	224
	HDD18.3	3152	537	465	380	283	133	77	27	31	104	235	399	480
	CDD10.0	1180	1	2	10	28	139	201	284	291	158	59	4	2
	CDD18.3	160	0	0	0	0	4	28	52	64	12	1	0	0
	CDH23.3	1183	0	0	1	2	38	193	382	485	74	8	0	0
	CDH26.7	281	0	0	0	0	1	36	91	144	8	2	0	0
(14) Wind	WSAvg	2.9	3.0	3.2	3.5	3.1	2.8	2.7	2.8	2.7	2.6	2.7	2.9	3.0
(15) Precipitation	PrecAvg	664	38	35	38	48	81	90	72	73	50	44	55	45
	PrecMax	976	93	103	138	119	212	181	165	168	137	115	104	115
	PrecMin	454	4	9	7	12	25	27	7	13	19	4	9	1
	PrecStd	110	19	19	23	28	36	39	39	35	23	28	25	25
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.1	16.6	21.3	22.4	26.2	29.8	31.3	33.1	27.9	23.8	15.9	14.1
		MCWB	9.0	10.2	12.7	13.0	16.7	20.2	20.1	20.2	17.3	15.8	11.3	10.4
	2%	DB	10.1	13.0	16.8	19.6	24.4	27.5	29.1	30.3	25.6	20.4	13.6	12.3
		MCWB	7.2	8.8	10.3	11.8	15.2	18.9	19.6	19.4	17.5	14.8	10.7	9.2
	5%	DB	8.6	10.6	14.0	17.3	22.7	25.6	27.3	28.0	23.3	17.8	12.0	10.4
		MCWB	6.2	7.2	8.8	10.7	14.4	18.0	18.8	18.8	16.7	13.6	9.6	8.0
	10%	DB	7.5	8.6	12.2	15.1	21.0	23.6	25.4	26.0	21.1	15.9	10.5	8.6
		MCWB	5.4	6.2	8.0	9.5	14.0	16.9	17.9	18.0	15.8	12.7	8.3	6.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.4	10.5	12.8	13.5	18.1	21.4	21.5	21.4	20.0	16.6	12.2	11.2
		MCDB	12.2	15.3	20.6	20.5	23.4	27.5	28.2	29.8	24.9	21.6	14.6	13.2
	2%	WB	7.8	9.1	10.7	12.4	16.4	20.0	20.5	20.4	18.2	15.4	11.1	9.5
		MCDB	9.9	12.9	15.6	18.5	21.8	26.0	26.8	28.2	22.9	19.4	13.1	11.8
	5%	WB	6.5	7.6	9.4	11.4	15.4	18.8	19.6	19.4	17.3	14.2	9.8	8.1
		MCDB	8.2	10.1	13.4	16.1	20.7	23.8	25.7	26.0	21.7	17.6	11.8	10.3
	10%	WB	5.6	6.4	8.4	10.3	14.5	17.6	18.6	18.6	16.3	13.0	8.3	6.8
		MCDB	7.3	8.7	11.8	14.4	19.7	22.6	24.1	24.6	20.6	15.3	10.2	8.5
(35) Mean Daily Temperature Range	5% DB	MDBR	4.9	6.3	7.1	7.8	8.8	8.0	8.7	9.3	8.4	7.4	5.6	4.4
		MCDBR	6.2	9.8	11.1	11.8	12.1	11.6	12.3	13.1	12.2	10.6	7.6	5.9
	5% WB	MCWBR	4.3	6.2	5.9	5.7	5.5	4.8	5.1	4.6	5.1	5.7	5.3	4.3
		MCWBR	5.4	8.6	9.7	10.1	10.1	10.5	10.6	11.2	10.5	9.9	6.8	5.8
(40) Clear-Sky Solar Irradiance	taub	0.308	0.333	0.364	0.398	0.408	0.413	0.413	0.409	0.382	0.357	0.329	0.305	0.305
		taud	2.442	2.384	2.323	2.255	2.258	2.279	2.271	2.304	2.358	2.420	2.453	2.442
	Ebn at Noon	757	815	844	848	853	850	843	828	815	774	721	708	708
		Edh at Noon	64	85	106	125	130	128	128	118	102	81	63	57
(44) All-Sky Solar Radiation	RadAvg	0.93	1.70	2.88	4.26	5.03	5.71	5.36	4.67	3.45	2.08	1.08	0.74	0.74
	RadStd	0.13	0.28	0.42	0.59	0.54	0.46	0.49	0.45	0.39	0.22	0.13	0.12	0.12

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (64 neighbors)	N/A	N/A	N/A	N/A	+0.34	+0.31	N/A	N/A	N/A	+20

Nomenclature: See separate page